FedDef: Robust Aggregation Against Model Poisoning Attacks in Federated Learning

Project Category: Application & Algorithmic (Combined)

Team Members

- Kunal V Kurundwadkar — IIT2024177
- Dewansh Khandelwal — IIT2024173
- Pratik Rakhecha — IIT2024176
- Kunal Khandelwal — IIT2024162

Motivation

Federated Learning (FL) allows multiple institutions, such as hospitals, to jointly train a shared model without exposing raw patient data, which makes it well suited to privacy-sensitive healthcare applications. However, standard aggregation schemes such as FedAvg implicitly trust every participating client, leaving the global model vulnerable to model-poisoning attacks in which a fraction of clients submit corrupted or adversarial parameter updates to degrade performance or embed malicious behaviour. Since even a small fraction of poisoned clients can substantially harm the shared model, a robust aggregation mechanism is essential before FL can be safely deployed in critical domains such as cardiovascular risk prediction.

Method

We use the Framingham Heart Study dataset to train a logistic-regression classifier that predicts ten-year coronary heart disease risk (TenYearCHD). The training data is partitioned across a configurable number of simulated clients, a tunable fraction of which behave maliciously by submitting randomized, out-of-distribution coefficient and intercept updates during federated averaging. We are building a defense layer that scores each client's per-round contribution using a z-score-based anomaly measure computed over client prediction deviations, accumulates a running confidence score per client, and down-weights or excludes updates whose deviation exceeds a threshold before the weighted-averaging step — rather than trusting every client in proportion to data size alone.

Intended Experiments

We will compare three settings over multiple training epochs: (1) FedAvg with no malicious clients as an upper-bound baseline, (2) FedAvg under attack with no defense, and (3) our anomaly-aware FedAvg defense. Evaluation will track test-accuracy convergence curves across epochs, vary the malicious-client fraction, and confirm that the defense does not degrade accuracy in the attack-free case. An interactive Streamlit interface lets us vary the number of clients, malicious fraction, learning rate, and epoch count for exploratory analysis.

Contributions

Kunal V Kurundwadkar: defense-mechanism design and z-score anomaly-scoring implementation. Dewansh Khandelwal: dataset preprocessing and the baseline FedAvg / logistic-regression pipeline. Pratik Rakhecha: malicious-client attack simulation and experiment orchestration. Kunal Khandelwal: Streamlit interface, result visualization, and report writing. All members will jointly contribute to evaluation, analysis, and the final report.